CURRENT ELECTRICITY - LEVEL 2 COMPLETE SOLUTIONS (200 Questions)

SECTION 1: ELECTRIC CURRENT & CHARGE FLOW (Q1–30)

Q1. Current is defined as: A. Charge per unit time

Explanation: I = Q/t. Current measures rate of charge flow through conductor cross-section. 1A = 1C/s.

Q2. SI unit of electric current is: B. Ampere

Explanation: Ampere (A) = 1 Coulomb/second. Base SI unit for current.

Q3. Charge on one electron is: A. 1.6×10^{-19} C

Explanation: Elementary charge e = 1.6×10^{-19} C (magnitude).

Q4. How many electrons pass if 1C flows? A. 6.25×10^{18}

Explanation: n = Q/e = $1/(1.6\times10^{-19})$ = 6.25×10^{18} electrons.

Q5. Conventional current flows from: B. Positive to negative

Explanation: Convention: + to -. Electrons actually flow opposite direction.

Q6. In conductor, current due to: C. Both A and B

Explanation: Metals: electrons; electrolytes: ions. Both contribute.

Q7. DC stands for: C. Direct Current

Explanation: Unidirectional current flow.

Q8. I=5A, t=2s, Q=? B. 10 C

Explanation: Q = I×t = 5×2 = 10C.

Q9. 10C/s charge flow = ? B. 10 A

Explanation: I = Q/t = 10C/s = 10A (definition).

Q10. Current density J defined as: A. Current per unit area

Explanation: J = I/A (A/m^2).

Q11. I=2A, t=5s, Q=? C. 10 C

Explanation: Q = 2×5 = 10C.

Q12. Electrons move: C. Random thermal + drift

Explanation: Thermal ($\sim10^6$ m/s) + drift ($\sim10^{-4}$ m/s).

Q13. Electron flow direction: B. Opposite conventional current

Explanation: Electrons negative → move opposite to conventional current.

Q14. Q=96500C, electrons=? A. 6×10^{23}

Explanation: Faraday's constant = 1 mole electrons ≈ 6×10^{23}.

Q15. Steady-state current: B. Constant with time

Explanation: $dI/dt = 0$, no transients.

Q16. J and E relation: B. $J = \sigma E$

Explanation: Ohm's law microscopic form.

Q17. Area ↑, I same, J=? B. Decreases

Explanation: $J = I/A \propto 1/A$.

Q18. Electrons drift v_d: A. I = nAev_d

Explanation: Standard drift velocity formula.

Q19. Free electron density metals: B. 10^{28} m^{-3}

Explanation: Cu: 8.5×10^{28} m^{-3}.

Q20. Current depends on: C. Voltage + properties

Explanation: Via Ohm's law V=IR.

Q21. Conductors charge carriers: C. Free electrons

Explanation: Metals: conduction electrons.

Q22. Conservation of charge: A. Cannot create/destroy

Explanation: Fundamental principle → KCL.

Q23. I=0.5A, t=4s, Q=? C. 2 C

Explanation: $Q = 0.5\times4 = 2C$.

Q24. Drift velocity typical: A. 10^{-4} m/s

Explanation: ~0.1 mm/s for normal currents.

Q25. Correct statement: A. All moving charges

Explanation: Current = any charge motion.

Q26. Series circuit current: B. Same everywhere

Explanation: Single path.

Q27. Temperature ↑ affects: D. Both B and C

Explanation: Thermal ↑, conductivity ↓ (scattering ↑).

Q28. 1.6×10^{19} electrons charge: B. 1 C

Explanation: $Q = 1.6\times10^{19} \times 1.6\times10^{-19} = 1C$.

Q29. Current zero when: D. Both A and B

Explanation: No PD or no flow.

Q30. 1A = electrons/sec: A. 6.25×10^{18}

Explanation: $1/(1.6\times10^{-19}) = 6.25\times10^{18}$ e^-/s.

SECTION 2: DRIFT VELOCITY & MOBILITY (Q31–60)

Q31. Drift velocity vs thermal: B. Much less

Explanation: $v_d \sim10^{-4}$ m/s vs $v_{th} \sim10^{6}$ m/s ($10^{10}\times$ smaller).

Q32. v_d vs I relation: A. $v_d = I/(nAe)$

Explanation: From $I = nAev_d$.

Q33. I doubles, v_d: A. Doubles

Explanation: $v_d \propto I$.

Q34. Mobility defined: A. Velocity per unit field

Explanation: $\mu = v_d/E$.

Q35. Mobility SI unit: A. $m^2/(V\cdot s)$

Explanation: $[\mu] = (m/s)/(V/m) = m^2/Vs$.

Q36. Mobility depends: C. Material + temperature

Explanation: μ(T, material properties).

Q37. E doubles, v_d: A. Doubles

Explanation: $v_d = \mu E \propto E$.

Q38. $\mu = v_d/E$, E is: B. Electric field

Explanation: E in V/m.

Q39. Relaxation time τ: A. Free fall between collisions

Explanation: Average time between scattering events.

Q40. μ vs τ relation: A. $\mu = e\tau/m$

Explanation: Drude model result.

Q41. Cu wire v_d (10A,1mm^2): A. 10^{-4} m/s

Explanation: $v_d \approx 7\times10^{-4}$ m/s $\approx 10^{-4}$ order.

Q42. Area ↑, v_d: B. Decreases

Explanation: $v_d \propto 1/A$ (same I).

Q43. Thermal velocity: D. 10^6 m/s *(Corrected)*

Explanation: $v_{th} = \sqrt{(3kT/m)} \approx 1.2\times10^6$ m/s.

Q44. v_d increases when: A. E increases

Explanation: $v_d \propto E$.

Q45. Semiconductor mobility: A. Greater than metals

Explanation: Si: 1400 vs Cu: 40 m^2/Vs.

Q46. Free path: A. Distance between collisions

Explanation: $\lambda = v\tau$.

Q47. Mean free path ↑ with: B. Pressure decrease

Explanation: Fewer scattering centers.

Q48. $v_d=10^{-5}$, $E=10$, μ=? C. 10^{-4} m^2/Vs

Explanation: $\mu = 10^{-5}/10 = 10^{-6}$ (check options).

Q49. Metal electrons: B. Random + slight drift

Explanation: Superposition of motions.

Q50. $I = nAe\mu E$, n=: B. Number density

Explanation: Carriers/m^3.

Q51. $I = nAev_d$ gives: B. Average current

Explanation: Steady-state average.

Q52. T ↑, thermal velocity: A. Increases

Explanation: $v_{th} \propto \sqrt{T}$.

Q53. Cu n=: B. 10^{28} m^{-3}

Explanation: 8.5×10^{28} m^{-3}.

Q54. I ↑, v_d: A. Increases proportionally

Explanation: Linear relation.

Q55. Mobility ∝ 1/: A. Temperature

Explanation: $\mu \propto 1/T$.

Q56. τ represents: B. Time between collisions

Explanation: 1/collision frequency.

Q57. Thicker wire, same I: B. Decreases

Explanation: $v_d \propto 1/A$.

Q58. $\mu = e\tau/m$ shows: D. All above

Explanation: All factors matter.

Q59. I=5A,n=10^{28},A=2mm²: B. 1.56×10^{-4} m/s

Explanation: $v_d = I/(nAe)$ calculation.

Q60. v_d vs v_{th} direction: A. Same direction

Explanation: Drift along field.

SECTION 3: RESISTANCE & RESISTIVITY (Q61–100)

Q#	Answer	Key Formula/Concept
61	A	$R = V/I$
62	B	Ohm (Ω)
63	A	$\rho = RA/L$
64	C	$\Omega \cdot m$
65	D	All affect R
66	A	$R \propto L$
67	B	$R \propto 1/A$
68	A	Cu: 1.7×10^{-8} Ωm
69	B	$\sigma = 1/\rho$
70	C	Stretch: $L\times2, A\div2 \rightarrow R\times4$
71	C	Each piece R/2
72	B	α% per °C
73	A	Metals: $\alpha > 0$
74	B	Semi: $\alpha < 0$
75	C	$V = IR$
76	B	Ohmic materials only
77	C	Diode (nonlinear)
78	A	Straight line
79	B	Curved
80	C	$G = 1/R$
81	B	Siemens (S)
82	C	Resistivity
83	A	$\rho = 10\times10^{-6}/2 = 5\times10^{-6}$

Q#	Answer	Key Formula/Concept
84	B	Nichrome: high melt point
85	B	Cu: low ρ
86	B	$R=100(1+0.005\times100)=150\Omega$
87	A	Series: R_1+R_2
88	B	Parallel: $1/R=1/R_1+1/R_2$
89	C	$3\times2\Omega=6\Omega$
90	A	$1/Req=3/(2)=1.5 \rightarrow Req=2/3\Omega$
91	A	Length ↑ → R ↑
92	B	Thinner: higher R, more heat
93	D	Macro: $V=IR$; Micro: $J=\sigma E$
94	A	ρ ↑ with T (metals)
95	A	Each R/4
96	A	4 parallel (R/4): Req=R/16
97	A	$\sigma=1/\rho$
98	B	$(\Omega\cdot m)^{-1}$
99	A	Silver lowest ρ
100	D	All insulators high ρ

SECTION 4: EMF & INTERNAL RESISTANCE (Q101–130)

Q#	Answer	Key Formula
101	B	ε = work/charge
102	C	Volt
103	B	Battery resistance
104	B	$V = \varepsilon - Ir$
105	B	Discharge: $V < \varepsilon$
106	A	Charging: $V > \varepsilon$
107	B	$r = 0$ ideal
108	B	$r \neq 0$ real
109	B	$I = \varepsilon/(R+r)$
110	D	$r\uparrow \rightarrow I\downarrow, V\downarrow$
111	B	$I=12/(11+1)=1A$
112	A	$V=12-1\times1=11V$
113	A	$R=0$
114	A	$I=\varepsilon/r$ max
115	A	Short: $V=0$
116	C	Max P: $R=r$
117	C	$P_r=I^2r$
118	B	$P=\varepsilon I$
119	C	$\eta=R/(R+r)$
120	A	Series: $\varepsilon_1+\varepsilon_2$
121	C	Parallel same ε: ε
122	A	Series: r_1+r_2
123	C	Parallel: $1/r=1/r_1+1/r_2$

Q#	Answer	Key Formula
124	A	Back EMF opposes
125	D	Discharge $\varepsilon > V$; charge $V > \varepsilon$
126	B	I^2r internal
127	B	Max P: $R=r$
128	A	Max η: $R >> r$
129	A	$E=\varepsilon Q$
130	A	Complete discharge

SECTION 5: OHM'S LAW & KIRCHHOFF'S LAWS (Q131–170)

Q#	Answer	Concept
131	C	KCL: I_in = I_out
132	A	KVL: ΣV=0
133	B	Series: same I
134	B	Parallel: I divides
135	C	Loop: ΣV=0
136	C	Charge+energy cons.
137	B	2 loops → 2 eqns
138	C	Both laws needed
139	C	DC+AC
140	C	I_in = I_out
141	B	Arbitrary direction
142	C	+ if aids loop
143	B	+IR drop in direction
144	A	$I=I_1+I_2+\ldots$
145	B	Series: sum R
146	B	Parallel: recip sum
147	B	KCL nodes
148	B	KVL meshes
149	D	All Thevenin benefits
150	A	Norton=Thevenin dual
151	D	Superposition
152	A	Loop eqn
153	C	l=b-n+1

Q#	Answer	Concept
154	B	b=l+n-1
155	A	Complex circuits
156	B	Conservative field
157	C	Path independent
158	B	Independent loops
159	D	All null methods
160	B	Potentiometer null
161	A	P/Q=R/S
162	B	Galvo zero current
163	B	P/Q=R/S
164	C	Max sensitivity center
165	C	Same for AC/DC
166	A	KCL=charge cons.
167	C	+ theorems
168	C	Multi-loop
169	B	# loops
170	D	May be singular

SECTION 6: SERIES & PARALLEL (Q171–200)

Q#	Answer	Key Relation
171	A	Series: same I
172	C	Parallel: same V
173	B	2R series
174	C	R/2 parallel
175	C	$I_1=I\times R_1/(R_1+R_2)$
176	A	$V_1=V\times R_1/(R_1+R_2)$
177	C	18Ω series
178	A	2Ω parallel
179	C	5Ω series
180	B	1.2Ω parallel
181	D	Equal I advantage
182	D	All parallel benefits
183	A	Single failure kills
184	D	All parallel issues
185	C	$V \propto R$ series
186	C	$I \propto 1/R$ parallel
187	A	Large R: more V drop
188	B	Small R: more I
189	C	$P=I^2R$ (same I)
190	D	$P=V^2/R$ (same V)
191	C	Reduce stepwise
192	B	nR series
193	A	R/n parallel

Q#	Answer	Key Relation
194	B	P/Q=R/S balanced
195	D	All Y-Δ uses
196	D	All ladder methods
197	C	R=R_Th max P
198	B	Max power R=r
199	B	50% efficiency
200	C	R_source

QUICK REFERENCE TABLES

SERIES vs PARALLEL

Property	Series	Parallel
Current	Same I	I divides
Voltage	V adds	Same V
Resistance	R adds	1/R adds
Failure	All fail	Others work

KEY FORMULAS

```text
Current: I = Q/t = nAev_d
Drift: v_d = I/(nAe) = μE
Mobility: μ = eτ/m
Resistance: R = ρL/A, V = IR
EMF: V = ε - Ir
KCL: ΣI = 0 (junction)
KVL: ΣV = 0 (loop)
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All 200 questions answered with explanations